ECOCYCLE

CIRCULAR E-WASTE COLLECTION & SUSTAINABILITY PLATFORM

v1.0.0 · UI v2.4 Pro

Complete Project Documentation & Technical Specification

End-to-end full-stack architecture, database schemas, REST API reference, environmental impact engine, interactive Leaflet GIS map, and glassmorphic UI design system.

Express 5
Node.js Backend
Mongoose 8 & JWT

14 Categories
Waste Taxonomy
Weight & CO2 Factors

16 Verified Hubs
India Network
42 City Coordinates

8-Stage Audit
Request Lifecycle
Pending to Completed

Project Identity & Environment Summary

PROPERTY	SPECIFICATION
Package Name	ecocycle-ewaste-management (v1.0.0, ISC License)
Entry Point & Server	server.js (Express 5.3.0, default PORT=5000 with auto port-conflict fallback)
Primary Database	MongoDB (Mongoose v8.24.8) with resilient in-memory & localStorage offline fallback
Reference SQL Schema	DATABASE.sql (10 normalized MySQL tables + ER MODEL.png diagram)
Authentication	JWT (JSON Web Tokens, 7d expiry) + bcryptjs password hashing (10 salt rounds)
Mapping Engine	Locally bundled Leaflet JS v1.9.4 (zero external geocoding API dependency)
Frontend Stack	Vanilla ES6+ SPA (app.js), semantic HTML5 (index.html), Glassmorphic CSS3 (styles.css)
Document Generated	November 12, 2026

Built for Real-World Resilience

EcoCycle operates seamlessly both with a live MongoDB server and in zero-dependency offline preview mode —automatically falling back to in-memory datasets, local Leaflet assets, and browser storage.

01 Executive Summary, Personas & Demo Access

Overview of EcoCycle's mission, role-based access, and pre-seeded test credentials

EcoCycle is an end-to-end full-stack web platform engineered to streamline household and enterprise electronic waste (e-waste) collection, certified drop-off discovery, administrative logistics triage, and verified circular material recovery. It bridges citizens seeking responsible electronics disposal with authorized reclamation centers across India while quantifying real-world environmental impact.

User Personas & Role-Based Access Control (RBAC)

- **Citizen / Regular User (role: 'user'):** Registers or signs in to book home pickups or drop-off handovers across 14 e-waste categories, uses the live environmental impact estimator, explores certified recycling centers on an interactive map, tracks 8-stage request timelines, and can cancel pending requests.
- **Administrator / Operations Hub (role: 'admin'):** Views system-wide KPIs, status distribution donut charts, top category bars, and 6-week submission trends; performs 1-click Approve/Reject triage; transitions requests through the 8-stage lifecycle; deletes invalid records; and exports filtered datasets to formula-safe CSV.
- **Guest Map Explorer:** Can jump directly from the authentication screen into the interactive Leaflet map to locate verified recycling centers, electronics shops, and scrap yards across India without signing in.

Pre-Seeded Demo Accounts & Access Scopes

ROLE	NAME	EMAIL	PASSWORD	ACCESS SCOPE
User	Rahul Sharma	`rahul@example.com`	`demo1234`	Submit pickups/drop-offs, track own requests, map
User	Priya Verma	`priya@example.com`	`demo1234`	Submit collection requests, track own request history
Admin	System Admin	`admin@example.com`	`demo1234`	Approve/reject, 8-status pipeline, CSV export, analytics

Key Functional Highlights

- **Eco-Impact Sustainability Tracker:** Real-time calculation of total e-waste diverted (kg), CO2 emissions offset avoided (kg), precious metals recovered (Copper, Aluminum, Gold traces in grams/mg), and an animated Circularity Index progress bar.
- **6-Stage Collection Pipeline Funnel:** Visual pipeline widget tracking active items across Pending Review -> Approved -> Scheduled -> Collected -> Processing -> Completed, with click-to-filter navigation directly to the Requests view.
- **Live Impact Estimator & Safety Alerts:** As users choose a waste category and quantity (1-99) on the New Request form, EcoCycle dynamically computes estimated weight, CO2 savings, and recoverable metals while surfacing contextual safety warnings (e.g., swollen batteries, SIM/memory card removal).
- **Resilient Hybrid Persistence:** Connects to MongoDB via Mongoose ODM when available; if MongoDB or the server is unreachable, server routes and the frontend API client seamlessly fall back to in-memory datasets and browser localStorage without interrupting workflows.

02 System Architecture & Directory Structure

Full-stack technology manifest, runtime resilience, and repository organization

Technology Stack

LAYER	TECHNOLOGY / PACKAGE	VERSION / DETAILS	PURPOSE IN ECOCYCLE
Runtime	`Node.js`	`v18+ (v24 tested)`	Server-side JavaScript execution environment
Web Framework	`Express`	`^5.3.0`	REST API routing, JSON parsing, CORS, static hosting
Database / ODM	`MongoDB + Mongoose`	`mongodb ^6.20 / mongoose ^8.24.8`	Document persistence, schema validation, hooks
Authentication	`jsonwebtoken + bcryptjs`	`jwt ^9.0.3 / bcryptjs ^3.0.3`	Stateless Bearer JWT auth (7d) & password hashing
Configuration	`dotenv`	`^17.4.2`	Loads PORT, MONGODB_URI, JWT_SECRET from .env
Mapping / GIS	`Leaflet JS`	`v1.9.4 (local vendor + CDN)`	Interactive map markers, circle layers, popups
Frontend UI	`HTML5, CSS3, ES6 JS`	`SPA Hash Router (#/...)`	Glassmorphic UI, dark/light themes, SVG sprite

Project File & Folder Manifest

PATH	TYPE	RESPONSIBILITY
`server.js`	**Backend**	Express 5 app, MongoDB connection, auto-seeder, health API, auto port fallback
`middleware/auth.js`	**Middleware**	JWT generation (generateToken), authenticate & requireAdmin guards
`middleware/errorHandler.js`	**Middleware**	Handles Mongoose ValidationError, duplicate key 11000, and 500 errors
`models/User.js`	**Schema**	User accounts, role enum, bcrypt pre-save hashing, comparePassword()
`models/CollectionRequest.js`	**Schema**	Requests, numericId, 8-state status enum, embedded historySchema array
`models/WasteCategory.js`	**Schema**	14 e-waste categories, avgWeightPerUnit, carbonOffsetPerKg, metals
`models/RecyclingCenter.js`	**Schema**	Drop-off hubs with lat/lng, verifiedPartner, acceptedCategories, hours
`models/RecyclingRecord.js`	**Schema**	Post-collection weight audit & recoveredMaterials breakdown
`routes/auth.js`	**API Route**	POST /login, POST /register, GET /session, POST /logout + memory fallback
`routes/requests.js`	**API Route**	CRUD, status/method/search filtering, RBAC enforcement, audit history
`routes/categories.js`	**API Route**	GET /api/categories with auto-seed of 14 default categories
`routes/centers.js`	**API Route**	GET /api/centers with auto-seed of 16 verified Indian recycling hubs
`routes/stats.js`	**API Route**	GET /api/stats computing CO2 saved, Copper/Aluminum/Gold & counts

PATH	TYPE	RESPONSIBILITY
`scripts/seed.js`	**CLI Script**	npm run seed: populates demo users, categories, centers & sample requests
`index.html / styles.css / app.js`	**Frontend**	Single-page application UI, glassmorphic themes, charts, drawer & map
`DATABASE.sql & ER MODEL.png`	**Reference**	10-table MySQL relational reference schema & visual Entity-Relationship diagram
`vendor/leaflet/ & fonts/ & Assest/`	**Assets**	Offline Leaflet engine, IBM Plex Sans woff2 fonts, EcoCycle brand logo

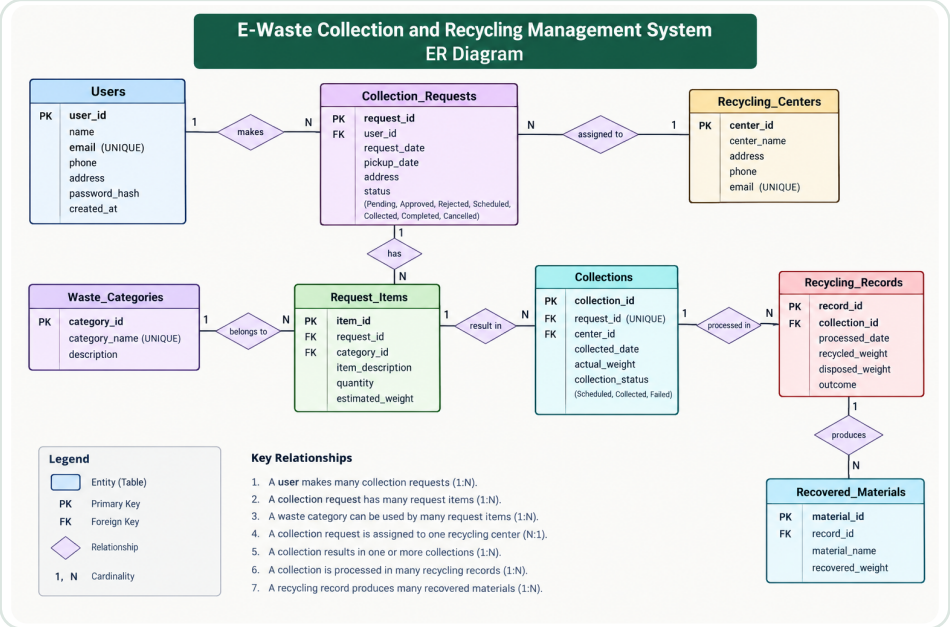
macOS AirPlay Port Conflict & Auto-Seeding Behavior

server.js listens on process.env.PORT || 5000. On macOS, ControlCenter (AirPlay Receiver) often occupies port 5000; server.js detects EADDRINUSE and automatically increments to port 5001 (or next available port). On successful MongoDB connection, autoSeedIfEmpty() automatically seeds admin@example.com and rahul@example.com if the Users collection is empty.

03 Database Architecture & ER Model

MongoDB Mongoose ODM collections alongside the 10-table MySQL reference schema

Entity-Relationship Diagram (ER MODEL.png)



MongoDB Mongoose Collections (models/)

MODEL	KEY FIELDS & INDEXES	HOOKS & TRANSFORMATIONS
`User`	name, email (unique, lowercase), password (min 6), role ['user','admin'], phone, address	Pre-save bcrypt salt(10) hash; toJSON strips password & __v, maps id=_id
`CollectionRequest`	numericId (unique index), user, email (index), userId (ref User), category, method ['Pickup','Drop-off'], qty (1..99), date, location, status (8 enums, index), estimatedWeight, actualWeight, history[]	Pre-save assigns numericId = Date.now() and initial 'Pending' history entry; toJSON exposes id = numericId

MODEL	KEY FIELDS & INDEXES	HOOKS & TRANSFORMATIONS
`WasteCategory`	name (unique), code (uppercase), desc, note, avgWeightPerUnit (default 2.5 kg), carbonOffsetPerKg (1.4), preciousMetals[]	Stores unit weights & environmental factors used in impact calculation
`RecyclingCenter`	name, address, city (default 'Jaipur'), phone, email, lat, lng, verifiedPartner, acceptedCategories[], openingHours	Provides geospatial coordinates for Leaflet map pins & drop-off selection
`RecyclingRecord`	collectionRequestId (ref), centerId (ref), processedDate, totalWeightKg, recycledWeightKg, disposedWeightKg, outcome, recoveredMaterials[]	Tracks post-processing material breakdown (materialName, weightKg)

Relational SQL Reference Schema (DATABASE.sql)

For relational deployments, DATABASE.sql defines the ewaste_management database with 10 normalized tables: Users, Admins, Waste_Categories, Recycling_Centers, Collection_Requests, Request_Status_History (cascade delete), Request_Items (quantity > 0 checks), Collections, Recycling_Records, and Recovered_Materials, complete with foreign keys, composite indexes, and sample seed rows.

04 REST API Reference & Security Architecture

Authenticated endpoints, role enforcement, and input/output safeguards

Complete REST API Endpoints

METHOD	ENDPOINT	AUTH / ROLE	DESCRIPTION & BEHAVIOR
`GET`	`/api/health`	**Public**	Returns service status, MongoDB readyState, masked URI, uptime
`POST`	`/api/auth/register`	**Public**	Creates 'user' account (name, valid email, 8-128 char password), returns JWT
`POST`	`/api/auth/login`	**Public**	Verifies email, password & role against MongoDB or demo users, returns JWT
`GET`	`/api/auth/session`	**Bearer JWT**	Validates token and returns authenticated user profile
`POST`	`/api/auth/logout`	**Public**	Acknowledges sign-out; client clears ecocycle_jwt_token
`GET`	`/api/requests`	**Bearer JWT**	Lists requests (user: own only; admin: all). Supports ?status, ?method, ?search
`POST`	`/api/requests`	**Bearer JWT**	Creates request (qty 1..99), looks up category avgWeightPerUnit, status='Pending'
`GET`	`/api/requests/:id`	**Bearer JWT**	Fetches single request by numericId or _id; enforces owner or admin check
`PATCH`	`/api/requests/:id/status`	**Bearer JWT**	Updates status & appends history[]. Non-admins can only cancel own Pending
`DELETE`	`/api/requests/:id`	**Admin Only**	Permanently deletes request from MongoDB & fallback store (requireAdmin)
`GET`	`/api/categories`	**Public**	Lists 14 waste categories sorted by name (auto-seeds DB if empty)

METHOD	ENDPOINT	AUTH / ROLE	DESCRIPTION & BEHAVIOR
GET	/api/centers	**Public**	Lists 16 verified recycling centers with coordinates (auto-seeds DB if empty)
GET	/api/stats	**Bearer JWT**	Computes counts, total/completed kg, CO2 saved, Copper/Aluminum/Gold

Security & Data Integrity Safeguards

- **JWT & Password Hashing:** Tokens are signed with JWT_SECRET (default 7d expiry) and verified by authenticate middleware. Passwords are hashed via bcryptjs with 10 salt rounds in User.js pre-save hook.
- **Strict Role Separation:** Public registration (/api/auth/register) always forces role: 'user'. Administrator endpoints (DELETE /api/requests/:id and status progressions) require role === 'admin' via requireAdmin or route checks.
- **CSV Formula Injection Prevention:** In app.js exportCsv(), any cell starting with =, +, -, @, \t, or \r is prefixed with a single quote (') before RFC-4180 quoting and UTF-8 BOM export.
- **XSS Output Escaping:** Frontend rendering escapes dynamic fields with esc() before injecting HTML into tables, drawers, timelines, and Leaflet popups.

05 E-Waste Categories, Impact Engine & Hub Network

14 supported e-waste streams, sustainability formulas, and verified Indian recycling centers

14 Accepted E-Waste Categories (routes/categories.js & app.js)

CODE	CATEGORY	AVG WT	CO2/KG	INCLUDED ITEMS & SPECIAL HANDLING NOTE
PC	**Computers**	6.5 kg	1.8 kg	Desktops, laptops, monitors, keyboards, mice (Remove sensitive data)
PH	**Mobile phones**	0.3 kg	2.5 kg	Phones, tablets, chargers, accessories (Remove SIM & memory cards)
BT	**Batteries**	0.5 kg	1.2 kg	Rechargeable & household batteries (Keep damaged batteries separate)
TV	**Televisions**	14.0 kg	1.5 kg	LCD, LED, plasma & displays (Handle screens upright)
PR	**Printers**	7.0 kg	1.3 kg	Printers, scanners, fax, ink & toner (Remove loose cartridges)
CB	**Cables**	0.8 kg	2.1 kg	Power cords, adapters, USB cables & wiring (Bundle neatly)
GC	**Game consoles**	3.2 kg	1.6 kg	Consoles, controllers & handhelds (Sign out of accounts)
AU	**Audio equipment**	2.4 kg	1.4 kg	Speakers, headphones, radios, stereos (Keep accessories together)
CA	**Cameras**	0.9 kg	1.9 kg	Digital cameras, camcorders, lenses (Remove memory cards)
SA	**Small appliances**	2.2 kg	1.1 kg	Small kitchen, grooming & household appliances (Clean food containers)
NW	**Networking eq.**	1.1 kg	1.7 kg	Routers, modems, switches & network gear (Reset credentials)
WR	**Wearables**	0.2 kg	2.2 kg	Smartwatches, fitness trackers (Built-in batteries: do not open)
LT	**Lamps & lighting**	0.4 kg	1.0 kg	LED lamps & electronic lighting (Ask point about special handling)
OT	**Other small elec.**	0.6 kg	1.5 kg	Calculators, toys, power banks (Power banks: treat as batteries)

Environmental Impact Calculation Formulas (routes/stats.js & app.js)

- **CO2 Emissions Avoided:** completedWeightKg * 1.44 kg CO2 saved per kilogram of completed e-waste.
- **Precious & Base Metals Recovered:** Backend stats compute Copper (~120 g/kg), Aluminum (~140 g/kg), and Gold traces (~35 mg/kg) from completed weight; frontend overview estimates combined recoverable metals at ~135 g/kg (and 120 g/kg on completion certificates).
- **Circular Recovery Rate:** Percentage of total requests reaching 'Completed' status (defaults to 100% when zero requests exist).

Verified Partner Recycling Centers Network (16 Seeded Hubs)

The platform includes 16 pre-configured, verified partner hubs across 10 major Indian urban clusters: Jaipur (2 hubs: Malviya Industrial Area & VKIA), Delhi NCR (3 hubs: Gurugram Sector 18, Okhla Phase II, Noida Sector 62), Bengaluru (2 hubs: Peenya 2nd Stage & Whitefield EPIP), Mumbai (2 hubs: Andheri East MIDC & Navi Mumbai Kopar Khairane), Chennai (Guindy), Kolkata (Salt Lake Sector V), Hyderabad (HITEC City Madhapur), Pune (MIDC Bhosari), Ahmedabad (Vatva GIDC), Lucknow (Transport Nagar), and Chandigarh (Industrial Area Phase 1).

06 Frontend Views, UI/UX Design & Setup Guide

Single-page application views, glassmorphic design system, and deployment commands

Application Views & Interactive Components

ROUTE / VIEW	USER ROLE	IMPLEMENTED CAPABILITIES & UI COMPONENTS
`#auth (Sign In / Up)`	**Unauthenticated**	Split hero screen, role selector, password eye toggle, 1-click demo buttons, Guest Map link
`#/overview`	**User & Admin**	Time-of-day greeting, Quick Action cards, KPIs, Eco-Impact Tracker + Circularity bar, 6-stage Pipeline Funnel, Admin charts (Donut, Category Bars, 6-Week Columns, Review Queue)
`#/requests`	**User & Admin**	Status filter tabs with live badges, search/method/status filters, category icons, inline Cancel (User), formula-safe CSV export
`Detail Drawer`	**User & Admin**	Slide-in glass drawer, metadata list, Circularity Certificate (if Completed), reverse-chronological audit timeline, Admin status update & delete modal
`#/new`	**Regular User**	2-column form, live Impact Estimator, category safety tip banner, Pickup/ Drop-off toggle, quick date chips (Tomorrow/+3d/+7d)
`#/map`	**All + Guest**	Leaflet v1.9.4 map, 42-city local geocoder, 5/10/25 km radius, geolocation, type chips (Recycling/Shop/Scrap), Use for drop-off & Google Maps directions
`#/categories & #/guide`	**User & Admin**	14-category reference table with 1-click Request Pickup; 4-step safe recycling procedure & battery safety callout

Quick Start & Operations Commands

COMMAND / SETTING	PURPOSE
`npm install`	Install dependencies (bcryptjs, dotenv, express, jsonwebtoken, mongodb, mongoose)
`npm start / npm run dev`	Start Express 5 server + static SPA on http://localhost:5000 (or 5001 if 5000 is busy)

COMMAND / SETTING	PURPOSE
`npm run seed`	Execute scripts/seed.js to populate 3 users, 14 categories, 16 centers & 5 requests
`.env Configuration`	PORT=5000 | NODE_ENV=development | MONGODB_URI=mongodb://127.0.0.1:27017/ecocycle | JWT_SECRET=... | JWT_EXPIRES_IN=7d
`Keyboard Shortcuts`	Press '/' anywhere to focus Global Search; press 'Escape' to close drawers/modals/nav

UI Design Highlights (styles.css v15)

Built with Plus Jakarta Sans & Outfit typography, Nordic Forest Emerald accents, frosted glassmorphism surfaces (blur 20-28px), full Light & Dark mode persistence (ecocycle-theme in localStorage), prefers-reduced-motion accessibility support, skip-to-content link, and responsive breakpoints down to mobile (<580px).